Week 5 Lecture

Topic: Describing Processes and Sequences

Part 1: Grammar – Passive

Introduction

TED Talk – The solution to our energy problems… is a toaster

Preview activities:

🌐 Environmental & Energy Terms

- **Fossil Fuels** (noun)
  - *Definition:* Natural energy sources formed from the remains of ancient plants and animals (such as coal, oil, and natural gas).
  - *In this talk:* Most industrial manufacturers burn fossil fuels to create high-temperature heat.
- **Carbon Pollution / Emissions** (noun)
  - *Definition:* Greenhouse gases (mostly CO_2) released into the atmosphere that trap heat and damage the climate.
  - *In this talk:* Industrial heat is responsible for roughly 20% of global carbon pollution.
- **Intermittent** (adjective)
  - *Definition:* Not continuous; stopping and starting at irregular intervals.
  - *In this talk:* Renewable energy from wind and solar is intermittent because it depends on the weather (the wind blowing or the sun shining).

⚙️ Materials & Mechanical Terms

- **Composite** (noun)
  - *Definition:* A material made by combining two or more distinct elements to create a new substance with improved properties.
  - *In this talk:* A nickel and chromium composite was invented in 1905 to make better heating elements.
- **Filament / Heating Element** (noun)
  - *Definition:* A thin wire or metal piece that converts electricity into thermal energy (heat) due to electrical resistance.
  - *In this talk:* The large storage bricks are threaded with filaments, just like the ones inside a kitchen toaster.
- **Abundant** (adjective)
  - *Definition:* Available in large quantities; plentiful and easy to find.
  - *In this talk:* Materials like bricks, rocks, and molten salt are cheap and abundant, making them great for storing heat.

🌡️ Thermal & Structural Terms

- **Disperse** (verb)
  - *Definition:* To scatter, spread out, or distribute something evenly over a wide area.
  - *In this talk:* Coiled toaster wires allow heat to be dispersed evenly across a slice of bread or a structural brick.
- **Encased** (adjective/verb)
  - *Definition:* Completely enclosed, covered, or wrapped in a protective container or outer layer.
  - *In this talk:* The entire thermal battery system is encased in insulating steel.
- **Insulation** (noun)
  - *Definition:* A material used to trap heat, electricity, or sound and prevent it from escaping.
  - *In this talk:* Steel insulation ensures that less than 1% of the heat escapes the battery chamber daily.
- **Channels** (noun)
  - *Definition:* Narrow, open pathways or passages through which air, gas, or liquids can flow.
  - *In this talk:* The interlocking blocks are molded with open channels so that cold gases can be blown through them to collect the stored heat.

🧠 Quick Check: Preview Quiz

Before listening, fill in the blanks using words from the glossary above.

1. Solar power is highly dependent on the weather, making it an _____ energy source.
2. To keep the heat inside the system from escaping into the factory, the brick chamber must be heavily _____.
3. Concrete is a common building material made from a _____ of cement, sand, and gravel.
4. Cool air is forced through open _____ inside the hot bricks to absorb the thermal energy.

Listening for gist

Choose the best answer for each question.

1. **What is the main topic of this TED-Ed talk?**
   - A) The history of kitchen appliances and how they changed breakfast.
   - B) Using a century-old toaster technology to build large-scale heat batteries for industry.
   - C) The process of manufacturing high-quality bricks, rocks, and molten salt.
   - D) How wind and solar energy can completely replace the electric grid today.
2. **Why is industrial heat a major environmental problem according to the talk?**
   - A) It consumes over 80% of the world's fresh water supply.
   - B) It is too expensive for small companies to generate safely.
   - C) Most companies burn fossil fuels to achieve high temperatures, creating 20% of global carbon pollution.
   - D) Industrial factories release too much hot air directly into the atmosphere, causing global warming.

Language Reference:

passive (all tenses)

1 A lot of films **are shot** on location. 6.4
When **is** our car **being repaired**?
Artem's bike **has been stolen**.
The director died when the film **was being made**.
You**'ll be picked up** at the airport by one of our staff.
This bill **must be paid** tomorrow.
I love **being given** a massage.
2 The new concert hall **will be opened** by the King.
Napoleon **was directed** by Ridley Scott.

1 We often use the passive (*be* + past participle) when it isn't said, known, or important who does an action.

Artem's bike has been stolen. (= somebody has stolen Artem's bike, but we don't know who). In passive sentences, the object of the verb becomes the new subject.

2 If we want to say who did the action, we use *by*.

- We can often say things in two ways, in the active or in the passive. Compare:

  Napoleon *was directed by Ridley Scott.* (= we want to focus more on the film)

  Ridley Scott directed Napoleon. (= we want to focus more on the director)
- We form negatives and questions in the same way as in active sentences.

  Some films ***aren't shot*** *on location.*
  Is *your car* ***being repaired*** *today?*
- We often use the passive to talk about processes, e.g. scientific processes, and in formal language, such as news reports.

  Then the water ***is heated*** *to 100 degrees...*

  Many buildings in the city ***have been damaged*** *by the earthquake.*

Activity 1

Circle the correct form, active or passive.

The college *built* / *was built* in the 16th century.

1 The costumes for the show *are making* / *are being made* by hand.
2 The story *inspired* / *was inspired* him to make a film.
3 This castle *hasn't inhabited* / *hasn't been inhabited* for nearly a century.
4 His latest film *set* / *is set* in Vietnam in the 1960s.
5 The film *will shoot* / *will be shot* in the autumn.
6 The actors *aren't recording* / *aren't being recorded* the dialogue until next week.
7 The house *wasn't using* / *wasn't being used* by the owners during the winter.
8 The make-up artist *has transformed* / *has been transformed* the actor into a monster.
9 They *hadn't owned* / *hadn't been owned* the company for very long before they went bankrupt.
10 The photo *took* / *was taken* by my husband on the balcony of our hotel.

Activity 2

Complete with the passive so that the meaning is the same. Only use *by* if necessary.

People don't use this room very often.
This room isn't used very often.

1 They subtitle a lot of foreign films.
A lot of foreign films _____.

2 García Márquez wrote *Love in the Time of Cholera* in 1985.
Love in the Time of Cholera _____ in 1985.

3 Someone is repairing my laptop at the moment.
My laptop _____ at the moment.

4 They haven't released the DVD of the film yet.
The DVD of the film _____.

5 They won't finish the film until the spring.
The film _____ until the spring.

6 You have to collect the tickets from the box office.
The tickets _____ from the box office.

7 They hadn't told the actor about the changes in the script.
The actor _____ about the changes in the script.

8 Damien Chazelle directed *La La Land*.
La La Land _____.

9 They've already recorded the soundtrack.
The soundtrack _____.

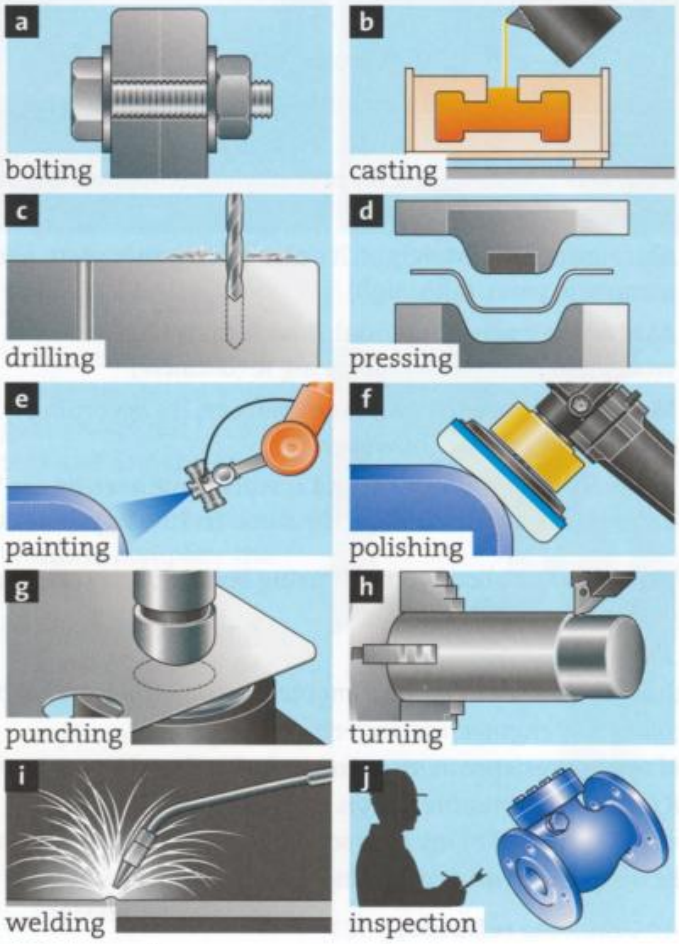
10 They were interviewing the director about the film.
The director _____ about the film.

Part 2: Listening and Oral Expression

Some manufacturing processes

1 Match each picture with a description.

1 A hole is made using a drill.
2 A surface is covered with paint.
3 A hole is punched in a material.
4 The work is checked carefully.
5 A surface is polished to make it smooth and shiny.
6 A material is pressed into the correct shape.
7 Hot liquid material is poured into a container and cooled to make the correct shape.
8 Two pieces are joined together using very high temperature.
9 A material is turned on a machine called a lathe. Part of the material is removed to make the correct shape.
10 Two pieces are joined together using a nut and a bolt.



Listening

A manufacturing process

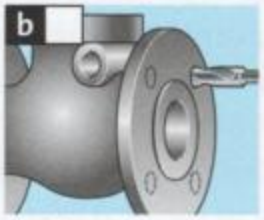
1 Look at the diagrams. Listen to the description of the manufacturing process for a check valve. Number the steps in the correct order.



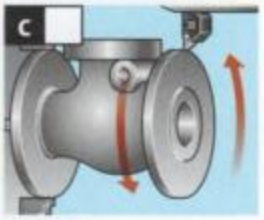
Check valve



bolting



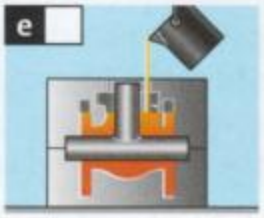
drilling



turning



inspection



casting



painting

2 Listen again. Answer the questions.

1 What verb tense does the speaker use to explain the process?

2 Why is this tense used?

3 What words does the speaker use to explain the order of the steps in the process?

Speaking
Explaining a manufacturing process

Work in pairs. Student A uses handout A, and student B uses handout B to explain a manufacturing process.

Part 3: Reading Comprehension

How Is Tea Made? Vocabulary and Reading Activities

Activity 1: Vocabulary

Before reading, match each word from the article (1–10) with its meaning (A–J). Write the letter in the Answer column.

No.	Word	Answer
1	pluck (plucked)	
2	moisture	
3	brew	
4	wither (withering)	
5	oxidation	
6	enzymes	
7	granules	
8	bruise	
9	grade	
10	tea chest	

Meanings

A. very small, rough pieces, like tiny grains

B. to make a hot drink by pouring hot water onto tea leaves

C. a large wooden box used to store and transport tea

D. a process in which something reacts with the air and changes colour

E. to pick leaves or flowers from a plant

F. to damage the surface of something by pressing or hitting it

G. to lose water and become soft and weak

H. natural chemicals in living things that cause changes

I. the small amount of water inside something

J. to put things in order by size, type or quality

Check in pairs: Use two of the words to say what you think happens to tea leaves in a factory. You will check your ideas when you read.

Activity 2: True or false?

Read the article. Write T (true) or F (false) for each statement. If a statement is false, correct it.

No.	Statement	T / F	Correction (if false)
1	Tea factories are usually far from the tea gardens.		
2	Only 24 hours pass between plucking the leaves and packing the tea.		
3	The withering stage takes only a few minutes.		
4	During oxidation, the leaves change colour from green to brown.		
5	In China, teas are always named after the person who made them.		
6	CTC tea was invented to increase the weight of tea that could be packed into a sack.		
7	Oolong tea is made from fully oxidised leaves.		
8	Green tea is made without the oxidation stage.		

Activity 3: Short answers

Answer each question in one or two full sentences, using your own words where you can.

1. Why are tea factories usually built close to the tea gardens?
   Answer: _____
2. Why are the leaves separated into batches of the same size?
   Answer: _____
3. What are the four stages of the orthodox method?
   Answer: _____
4. What happens to the leaves during oxidation, and what does this stage decide?
   Answer: _____
5. How is the CTC method different from the orthodox method of rolling?
   Answer: _____
6. Why does green tea have a light, fresh flavour?
   Answer: _____

Part 4: Describing a Process

Describing a process

A All the planets absorb heat from the sun in a **process** known as solar radiation. However, for life on our planet to continue, Earth must stay at an average surface temperature of around 15°C. For the Earth to stay at a **constant** temperature, it needs to absorb and emit radiation at the same rates.

As solar radiation travels to Earth, there is a **series** of **steps** which reduce the amount of radiation that reaches the surface:

- Firstly, some is **removed** as it enters the atmosphere – the mixture of gases that surround our planet. Around 25% is then reflected by the upper atmosphere and clouds, **while** a further 20% is absorbed by them.
- 5% of the radiation is reflected from the surface of the Earth – by deserts or ice.
- **Finally**, 50% is absorbed by the Earth's surface.
- Around 10% of this heat radiates from the surface, and the remaining 40% moves to the clouds and the atmosphere. This heat is then radiated back out into space, completing the cycle.

Source: Adapted from Johnson, K. (2016). *GCSE Physics For You.* (5th ed.). Oxford: Oxford University Press. p.214.

Glossary
absorb (*v*): to take in and keep heat, light, or other forms of energy
emit (*v*): to send out light or heat
radiate (*v*): to send out light, heat or energy
reflect (*v*): to throw back light or heat

Usage note sequence in a process

There are a number of different ways to talk about events in a process.
You can use adverbs: *firstly, then, next, after that, finally*

- ***Firstly**, some energy is removed by the clouds.*

You can also use words like *after, as, as soon as, once* and *when* to show how two events relate to each other:

- ***As soon as** sunlight hits the flower, its petals open.*

The phrase *at this stage* is used to help create a clear sequence:

- *The temperature reaches 100°C. **At this stage**, the water in the container begins to boil.*

Activity 1

1 Complete the student's summary of text A using sequencing words and phrases from the box.

as at this stage finally firstly then

[1]_____ solar radiation enters the Earth's atmosphere there are a number of steps that reduce it. [2]_____ , approximately 25% is reflected by clouds and the upper atmosphere. [3]_____ , a further 20% is absorbed by them. [4]_____ , 5% is reflected back from the Earth's surface. 50% is absorbed by the Earth's surface. 10% of this heat radiates directly from the surface and 40% is transferred back to the upper atmosphere and clouds. [5]_____ , this remaining amount radiates back into space.

Activity 2

2 Complete the sentences using words from the box.

constant process removed series steps

1 During the interview, he was asked a _____ of questions about his work experience.
2 Child psychologists are interested in the _____ of learning among children.
3 The number of students in higher education has remained _____ .
4 Following a large landslide, many trees and plants are _____ from the surface layer.
5 The discovery of DNA was one of the first _____ in understanding how to cure genetic conditions.

Important processes

B A blood culture **involves** taking a sample of blood from a patient and **transferring** the blood into a blood culture bottle. The bottle is then transported to the laboratory for processing. The **procedure** has three main **stages**: sample taking, transport and processing.

Glossary
culture (*n*): a group of cells grown in the laboratory for a medical or scientific study
sample (*n*): a small amount of a substance, often taken to be tested

C Many companies who use internet-based questionnaires now include 'progress indicators'. These tell the person who is completing the questionnaire how many questions they have answered. They also often say how much more information needs to be given to complete the **task**. **Previous** research suggested that the indicators could have a negative effect on how many people answer questionnaires. However, people who **regularly** answer these kinds of questionnaires report that if they know how long the process will be, they will answer more honestly and completely. It is also important that the **last** question in the process allows respondents to give their own opinions.

3 Read texts B and C. Match the content to the correct subject.

1 business studies __ 2 medicine __

4 Choose the correct word to complete the definitions.

1 *last / previous* – happening or existing before an event that you are talking about now
2 *stage / task* – a piece of work that has to be done
3 *procedure / stage* – a point, period or step in a process
4 *procedure / task* – a series of actions done in a particular order or way
5 *previously / regularly* – often
6 *involve / transfer* – to move from one place to another

5 Complete the sentences using the correct form of the highlighted words in texts B and C.

1 To stay fit and healthy it is important to get _____ exercise.
2 The final _____ of the experiment involved heating the materials to 1,000°C.
3 The _____ major change to the economy began with the banking crisis in 2008.
4 The current energy minister has _____ worked for a major oil and gas company.
5 The writing exam _____ describing a graph, and then writing an essay.
6 The oesophagus _____ food from the mouth to the stomach.

6 Write a short paragraph (two or three sentences) about one of the topics below. Try to use some of the words from the unit.

- Describe a process or cycle that you are familiar with.
- Explain which processes are important in your area of study.